

What is cost?

↳ An amount that has to be spent to obtain and/or use a commodity

↳ Cost means the cost to build a computer

↳ Price means the price to purchase a computer (Capital cost)

↳ Operational cost means the cost to operate the computers.

Operational cost:

↳ Closely linked to power consumption

↳ In warehouse-scale computing, which contains tens of thousands of servers, the cost to operate the computers is significant in addition to the cost of purchase

↳ Operational cost $\gg$ Capital cost

↳ The purchase of servers & networks is 60% of monthly cost to operate warehouse scale computing business

↳ About 40% is for power-use

Importance of understanding the trends

- Over the past 35 years, the use of technology improvements to lower cost, as well as increase performance, has been a major theme in the computer industry.
- Understanding **manufacturing cost** is essential for computer architects for making informed decisions about whether or not a new feature should be included where cost is an issue.
 - Imagine architects designing skyscrapers without any information on costs of steel beams and concrete
- Understanding **operational cost** is essential for lowering the cost to operate computers
 - Understanding **the trends in power and energy** are important for architects to design computers with low power and energy requirements (**Green Computing**)

Manufacturing cost:

him

↳ They are understood in terms of learning curve & cost

↳ Learning curve:

↳ The rate of a person's progress in gaining experience or new skills.

↳ In CA, the learning curve is the underlying principle used to understand the reduction in costs per unit as experience is gained while producing implementation technologies.

↳ Yield:

↳ The %age of manufactured devices that survive testing.
↳ The learning curve is best measured by change in the yield.

- Price per megabyte of DRAM has dropped over the long term.
- Microprocessors
 - Microprocessor prices drop over time
 - Price depends on volume
 - Volume is amount of a product manufactured in a given time
 - Increasing volumes decreases cost
 - Cost decreases by 10% for each doubling of volume

Calculating Manufacturing costs

$$\text{Cost of IC} = \frac{\text{Costs of \{Die + testing Die + Packaging and final test\}}}{\text{Final test Yield}}$$

$$\text{Cost of Die} = \frac{\text{Costs of Wafer}}{\text{Dies per Wafer} \times \text{Die Yield}}$$

$$\text{Dies per Wafer} = \frac{\pi(\text{Wafer diameter}/2)^2}{\text{Die Area}} - \frac{\pi(\text{Wafer diameter})}{\sqrt{2} \times \text{Die area}}$$

$$\text{Die Yield} = \frac{\text{Wafer yield} \times 1}{(1 + \text{Defects per unit area} \times \text{Die area})^N}$$

Defects per unit area = 0.016–0.047 defects per square cm

N = process-complexity factor = 10 - 14